Kenya transport and logistics frontend design

Research and product design specification | 17 October 2026 | Version 1.0

Recommended product direction

Build one connected transport platform with distinct customer, field-worker and business workspaces. Share identity, design components and trustworthy status information, while giving each role the tasks and permissions it needs. Customer simplicity should be supported by strong dispatch, custody and financial workflows behind the scenes.

This document translates the supplied Kenya Transport and Logistics Architecture into a frontend specification covering scheduled transport, rides, corporate travel, charter, courier, intercity parcels, freight, warehousing, returns and partner-operated journeys. It is intended for the product owner, designer, frontend engineers and operational leads. [A]

What the design should achieve

• Customers understand the total price, service promise, pickup location and next step before committing.

• Drivers can assess offers, complete assigned work, report problems and understand earnings without unnecessary typing.

• Operators can identify blocked work, manage capacity and reconcile what they earned with what they received.

• Conductors, branch teams, finance and support can finish the handoffs that make the customer promise possible.

How to use the document

Pages 2 to 4 explain the research, people and service coverage. Pages 5 to 16 specify workspaces and illustrate key screens. Pages 17 to 21 cover cross-role journeys, visual rules, exceptional states and access. Page 22 defines delivery priorities and validation. Pages 23 and 24 contain linked sources.

Research findings are identified with numbered sources. Proposed screens, layouts, labels, policies and release priorities are recommendations for this platform. Illustrative names, amounts and counts in the layouts are fictional. The design preserves the architecture’s separation of commercial, operational and payment status. [A, pp. 18-19]

Research findings and design implications

Use established interaction patterns while preserving the requirements of Kenyan passenger and goods operations.

Reference	Publicly documented pattern	Recommended adaptation
Uber rider [1, 2]	Destination entry, pickup confirmation, arrival tracking, vehicle identity and optional PIN verification.	Keep a short booking journey and a prominent driver and vehicle identity card.
Uber driver [3, 4]	Upfront offer information in documented markets; trip and weekly earnings views.	Explain estimated net earnings, exclusions and adjustments before acceptance and after completion.
Bolt [5, 6]	Fleet portal for drivers, vehicles, reports and payouts; safety page lists sharing and pickup codes.	Separate operator management from driving; place safety tools within an active trip.
Bolt Business [7]	Book for another passenger, add travel notes and send ride details by SMS.	Support booker and traveler as separate people, with the right notifications for each.
BuuPass [8]	Bus search, seat selection and mobile-money payment are described in its booking guidance.	Use a departure-and-seat flow for scheduled travel with explicit boarding information.
Onfleet [9, 10]	Driver tasks, navigation, delivery evidence, customer tracking and barcode capture.	Use next-stop screens and package-level evidence, with recipient tracking without installation.
Uber Freight [11]	Load filters, load status and proof-of-delivery document upload.	Provide freight offers, accepted terms, milestones and document completeness.
Fleetio [12]	Inspections show open issues; passing a later inspection does not resolve an existing issue.	Show unresolved defects and require an authorised release before vehicle availability returns.
Odoo [13]	Barcode-based receipts and deliveries connect items to transfers and locations.	Use scan-first branch and warehouse workflows with discrepancy resolution.

These findings come from public product and help documentation, not access to private dashboards or company internals. Uber and Bolt describe market-dependent features; the presence of a feature on a public page does not establish availability for every Kenyan user. Do not copy their pricing rules, response promises or branding. [1, 6]

The people and their workspaces

Cover every participant without making everyone navigate the entire business.

Person	Workspace and first priority	Access boundary
Customer or sender	Customer mobile app and responsive web; book or follow an active order.	Own bookings, shipments, receipts and cases.
Recipient or guest traveler	Secure mobile web link; arrival details and verification.	Only the relevant shipment or journey.
Driver or courier	Dedicated field app; current job and next stop.	Assigned work, necessary contacts, own earnings.
Conductor	Assigned-departure mode; manifest and ticket scanner.	Current departure and authorised parcel manifest.
Operator or carrier owner	Responsive operator portal; readiness, work and statements.	Own organisation and authorised branches.
Dispatcher	Operations web workspace; unassigned and at-risk work.	Assigned organisational or regional scope.
Branch or hub agent	Scanner-friendly mobile or tablet interface; receive and hand over.	Assigned branch and manifests.
Warehouse picker or supervisor	Warehouse mode; pick tasks, bins and discrepancies.	Assigned stock owner, site and task scope.
Maintenance technician	Mobile work orders and inspections.	Assigned vehicles and repair records.
Corporate travel manager	Business portal; approvals, policies, rosters and invoices.	Own organisation and delegated cost centres.
Merchant or frequent shipper	Business shipping workspace; batches, stock and returns.	Own orders, inventory and billing.
Finance and approver	Finance workspace; unresolved collections and payout review.	Separate initiation and approval permissions.
Support and safety staff	Case workspace; priority and journey context.	Case-relevant records and controlled evidence.
Platform administrator	Administration; approvals, access and configuration.	Explicit cross-organisation privileges with audit.
Management and auditor	Read-only reports and evidence exports.	Approved aggregates or scoped audit records.

A person may hold several roles. Use an explicit organisation and workspace switcher that clears previous data and restores only authorised context. A menu switch never grants a role. [A, pp. 5-6]

Coverage across transport services

A shared account should lead to service-specific forms, fulfilment and evidence.

Service	Customer entry and core screens	Operational completion
Scheduled bus or shuttle	Travel; route, date, departure, seat, passenger, checkout, ticket.	Manifest, segment-aware boarding, arrival and disruption handling.
On-demand ride	Ride; pickup, destination, category, quote, matching, active trip.	Offer acceptance, pickup verification, start, finish and fare review.
Corporate transport	Business travel; policy check, cost centre, approval, roster.	Approved dispatch, employee access and monthly billing.
Charter and hire	Hire; itinerary, dates, capacity, quote, deposit and changes.	Reserve full vehicle interval, agreed extras and final settlement.
Local courier	Send; parties, locations, package details, quote and collection.	Collection proof, delivery verification and failed attempt.
Intercity parcels	Send; branch or door options, package labels and tracking.	Origin scan, line-haul manifest, destination scan and release.
Freight and trucking	Freight; load, equipment, access, time windows and quotes.	Carrier offer, dispatch, milestones, documents and charges.
Warehouse fulfilment	Business stock; inventory, replenishment, order and return.	Receive, bin, reserve, pick, pack, dispatch and reconcile.
Returns and claims	Order detail; reason, affected items, evidence and remedy.	Review, reverse shipment, inspection and authorised adjustment.
Cross-border and partner legs	Specialist request; route, partner reference and required documents.	Partner-controlled milestones, document exceptions and handoffs.

Enable only operationally supported services in the customer launcher. Cross-border, air, sea, rail and specialist cargo should initially use assisted requests and partner milestones where supported, rather than implying direct carrier booking. Restricted and temperature-controlled services remain unavailable until their operating procedures exist. [A, pp. 3, 11-12]

Passenger seats, baggage allowances and cargo capacity remain separate in forms and availability. A vehicle may carry approved mixed work, but parcel stops must not silently alter a published passenger timetable. [A, pp. 12-13]

Shared navigation and onboarding

Keep the product familiar across roles while making each workspace immediately useful.

Customer navigation

Use four mobile destinations: Home, Activity, Help and Account. Home offers Travel, Ride, Send and Hire; Freight sits under Send with a distinct form. Put the next active booking above promotions. Activity filters Upcoming, Active and Past across services. Business users gain a clearly labelled Business workspace after invitation.

Driver and staff navigation

The driver app uses Today, Jobs, Earnings and Help, with profile and documents in the account menu. Assigned trip screens take priority over the home map. Operator web uses Overview, Work, Fleet, People, Statements and Settings. Operations web groups Dispatch, Passenger, Goods, Branches, Fleet, Cases and Finance according to permission; avoid one enormous flat sidebar.

Account and document onboarding

• Allow customer route and price exploration before sign-in where feasible. Require identity and contact verification at the point needed to reserve or communicate.

• Operator onboarding asks for organisation details, service areas, payout details and the required documents. Show Draft, Submitted, Under review, Needs changes and Approved separately.

• Driver onboarding shows identity, licence, vehicle relationship and service eligibility as a checklist. An uploaded file is not an approved document.

• Invite staff into a named organisation and role. Explain the access being accepted. Use stronger sign-in and confirmation for privileged actions.

• Explain location, camera and notification permissions when the relevant task first needs them. Denying optional notifications must not block booking.

Forms and recovery

Use persistent field labels, saved drafts and step summaries. Validate phone numbers, dimensions, dates and passenger counts beside their fields. Preserve entered values after an error. Let users edit the pickup before payment without retyping the entire request. Show why a document was rejected and the exact replacement needed.

Every list needs loading, empty, no-results, failed-load and stale-data states. Empty screens should explain the first useful action, such as Add a vehicle or Search departures. A permission error should explain how to reach the organisation administrator without exposing another organisation’s data. [A, pp. 5-6, 21]

Customer travel and ride screens

Make booking confidence and pickup clarity the main measures of success.

C01 Home and service selection

Show a location or service area selector, active booking card, four service choices and recent routes. If location permission is denied, allow manual entry. Distinguish unavailable service from an empty search result. Customer help remains reachable without an active booking.

C02 Scheduled search and comparison

Collect origin, destination, travel date and passenger count. Results show operator, boarding stage, departure and estimated arrival, vehicle class, baggage allowance, available seats and total price for the party. Provide filters for departure time, operator and price. Label overnight arrival dates explicitly.

C03 Seat and passenger selection

Show vehicle orientation, seat labels and a legend for available, selected and unavailable seats. Provide an accessible list alternative. Calculate availability for the chosen route segments. Start the visible countdown only after the server creates the hold. If group seats cannot all be held, explain the alternatives before taking payment.

C04 Checkout and C05 Ticket

Review passenger, operator, boarding landmark, terms and price breakdown. Confirm the payer’s M-PESA number without asking for their M-PESA PIN. Payment processing and ticket confirmation appear as separate steps. A confirmed ticket includes booking reference, seat, boarding time, branch directions and offline-readable ticket code. Changes and cancellation show consequences before submission.

C06 Ride quote and matching

Confirm pickup pin, landmark and destination, then compare eligible vehicle categories by capacity, pickup estimate and price. After request, show matching progress with cancellation terms. If no driver accepts, offer retry or an available scheduled alternative. An accepted driver card shows photo, name, vehicle and plate. [1, 2]

C07 Active ride and C08 Completion

Keep arrival estimate, last update, pickup instructions, contact, trip sharing and safety access visible. Verify pickup with the agreed method. For a destination change, show the revised estimate and obtain customer acceptance before committing the change. Completion provides fare detail, payment status, receipt, feedback and a dispute path. Feedback is optional; the receipt is immediately accessible.

Design recommendation: expose assistance requirements during booking, but confirm actual operator capability before promising accessible boarding or specialist support. Include booking for someone else with separate payer and traveler contacts. [A, pp. 6, 9-10]

Customer goods freight and hire screens

Ask for the information that changes price or fulfilment, and explain why it is needed.

C09 Parcel request and quote

Choose branch-to-branch, door-to-door or supported mixed service. Collect sender and recipient contacts, pickup and delivery pin or branch, landmark, package count, weight, dimensions, declared value and handling needs. Use plain examples for size and weight. Explain whether a quote is final or subject to measurement and how changes will be approved.

C10 Shipment tracking and recipient access

Show a milestone timeline and package list, with an ETA window when justified. Display 2 of 3 delivered rather than marking the shipment complete. Open a map only when it adds useful last-mile detail. The recipient’s expiring link shows only their delivery, collection instructions and support; it must not reveal the sender’s history or other stops. [9; A, pp. 11, 16]

C11 Freight request and offer comparison

Collect equipment, load type, weight, volume, pickup and delivery windows, loading access and handling responsibility. Compare offers by approved carrier, total price, included services, timing and exclusions. Accepted terms remain available in the order. Label additional charges as proposed, approved or disputed; keep signed delivery documents separate from payment state. [11]

C12 Vehicle hire

Use a date and itinerary form with vehicle capacity, pickup point and included distance or hours. Quote review shows deposit, balance, cancellation terms and permitted overtime. Changes produce a revised quote and availability check. A quote request is not a confirmed vehicle reservation.

C13 Returns and claims

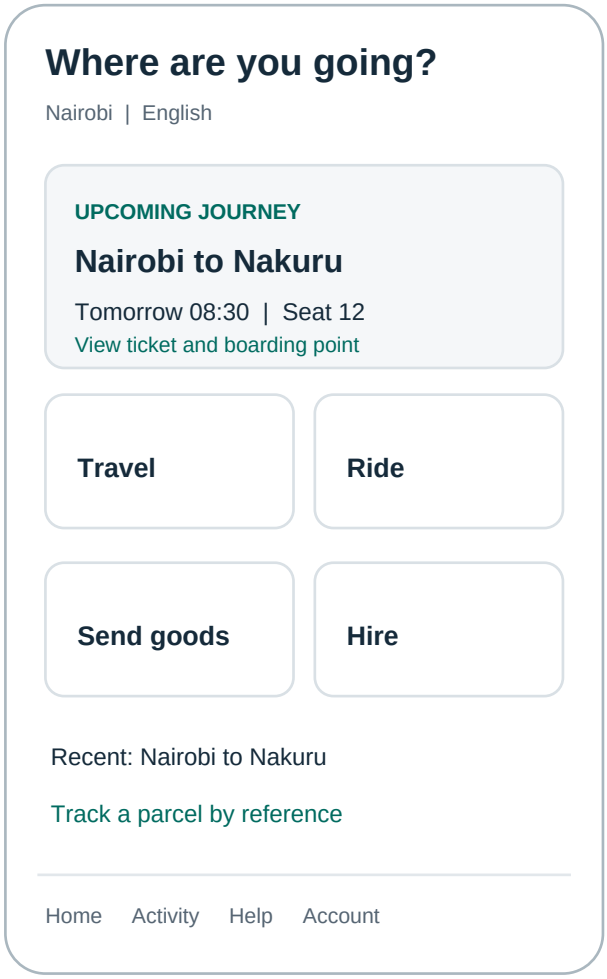
Start from the original order. Ask which package or service is affected, the issue, supporting evidence and requested resolution. Provide a case reference, next action and status timeline. An approved return creates a separate reverse shipment. A refund decision and its payment outcome remain separate.

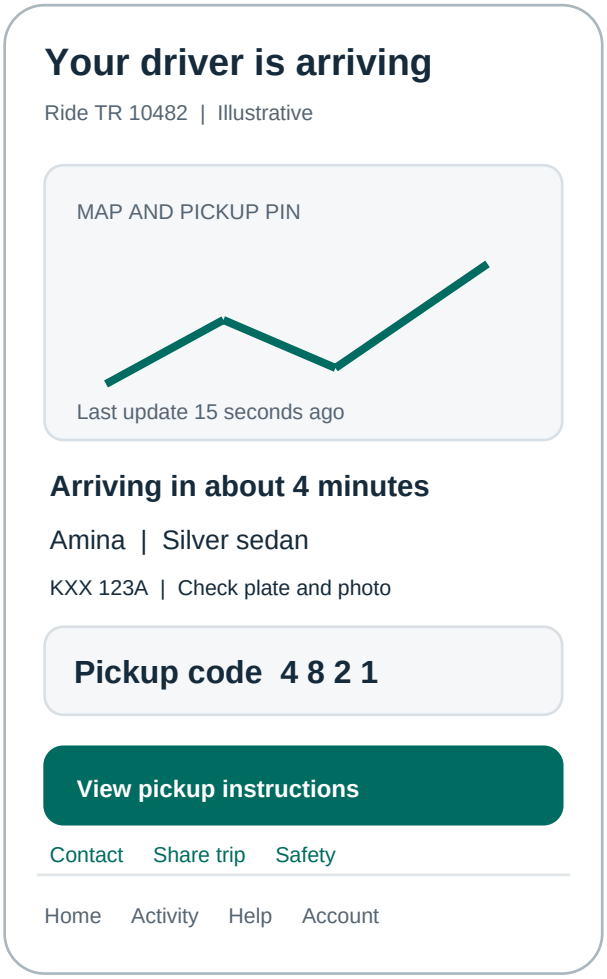
Business shipping and assisted customers

Frequent senders can save recipient addresses, duplicate a previous shipment and upload batches with a row-level preview of invalid records before submission. Branch agents can make assisted bookings with the customer’s contact and consent recorded. Send the customer their own reference and receipt through configured channels; the agent’s session must not become the customer’s account. [A, pp. 5-6, 11-12]

Customer screen concepts

Original layout proposals for a 390 pixel mobile viewport. The information hierarchy is the specification; sample values are illustrative.





Interaction notes

Home keeps existing commitments above new purchases. The active ride screen places identity and pickup verification near the arrival estimate. The map supports orientation; the text summary remains usable when maps or location updates fail.

The primary action changes with the journey. Before request it is Review ride; during matching it is View request; after assignment it becomes pickup guidance. Share trip and Help stay available without taking over the main action. Do not display a false live indicator when updates are stale.

Driver and courier experience

The driver should know the work, the earnings basis and the next safe action.

D01 Availability and offers

Show Online, Offline, On a job or Unavailable with a reason. Before going online, surface expiry or inspection blockers and a fix action. A time-limited offer shows service type, pickup distance and estimate, destination or service area, load or passenger needs, expected compensation and exclusions. Accept remains provisional until the server confirms the winning assignment. [3; A, p. 10]

D02 Today and D03 Active stop

Today lists assigned jobs by time with service badges, readiness and a durable sync indicator. Active stop prioritises location, landmark, recipient or passenger contact, required verification and one next action. Offer Open navigation in a supported mapping app. Use large controls and minimise interaction while moving.

D04 Verification and evidence

Passenger rides use pickup verification before start. Parcel collection requires expected package scans and condition evidence. Delivery verifies package and recipient with the configured OTP, signature or other approved evidence. Capture evidence per package, including partial delivery. A photo alone must not be treated as proof that the correct recipient received the correct item. [9, 10; A, p. 11]

D05 Earnings and cash

Show earnings by day, week and job, including gross amount, fees, approved adjustments, net amount and settlement status. Separate cash collected on behalf of the business from money owed to the driver. Explain pending adjustments and provide a dispute route. Avoid displaying estimates as paid income. [4; A, p. 15]

D06 Vehicle checks and D07 Help

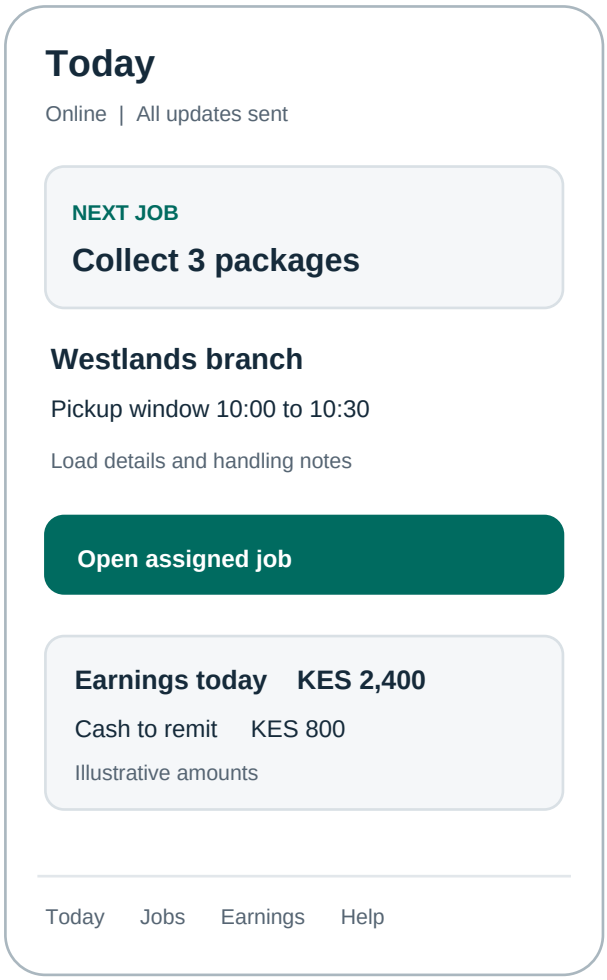
Provide pre-trip inspection, defect report, fuel receipt and incident capture where authorised. Safety-related defects create a visible availability block. Help offers operational assistance, incident reporting and payment questions with the job context already attached. A driver can report harassment and challenge an account restriction through a documented review path.

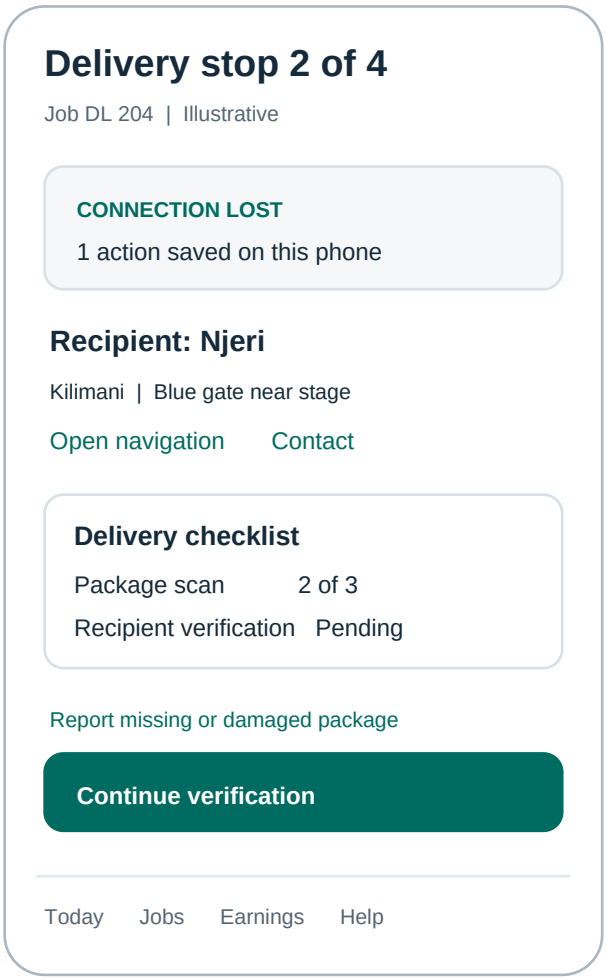
Offline work

Cache assigned work and required references securely. Queue evidence and task events with visible states: Saved on device, Waiting to send, Confirmed, Needs attention. Never label an offline capture as server-confirmed. Do not allow new assignments, payments or seat confirmations to be finalised locally. [A, p. 16]

Driver screen concepts

A task-led home and stop screen reduce navigation during field work.





Interaction notes

A driver working a scheduled departure sees a departure card; a courier sees the next stop. The bottom action advances only when the required checks pass. Delivery evidence has its own saved and uploaded status so a connection failure cannot erase completed field work.

The Earnings view separates cash to remit from settled income. Offer expiry is shown as text and a timer, never by colour alone. If another driver wins the assignment, replace the action with This job is no longer available and preserve the driver’s availability.

Conductor and branch workflows

Boarding and handover interfaces should optimise repeated verification, not general dashboard browsing.

B01 Departure and B02 Ticket scanner

Conductor mode opens on the assigned departure with route, vehicle, scheduled time, manifest freshness and boarding counts. Scan results distinguish Valid for this departure, Already boarded, Wrong departure, Cancelled and Needs online verification. Display passenger and segment details before recording boarding. Keep a manual reference lookup for camera failure.

Offline boarding must show when the manifest was downloaded and which decisions are pending. Multiple disconnected devices cannot promise immediate duplicate detection. Assign scanner responsibilities or reconciliation procedures explicitly, and flag conflicts after sync. Never hide a second boarding event. [A, pp. 9, 25]

B03 Counter booking and B04 Branch receipt

A counter agent searches departures and creates the same server-controlled seat hold as a customer. Payment evidence cannot be replaced with a free-form Paid checkbox. For parcels, select the incoming manifest, scan each label and record condition. Compare Expected, Received, Missing and Unexpected before closing receipt.

B05 Sort and load

Scan the package and destination zone or bin. Loading requires the correct vehicle manifest and checks package count, cargo weight and volume separately from seats. Highlight duplicate, wrong-route and unlisted packages. Exceptions require a reason and permitted supervisor resolution; they must not silently become normal receipts. [10, 13; A, p. 12]

B06 Release and handover

Show recipient or authorised delegate verification, package count and collection requirements. Do not reveal verification codes before the required recipient step. Record who handed over, who received, location and time. Display missing evidence or payment-policy blockers clearly. Reprinting a label preserves the existing package identity.

Physical workplace design

Use a scan-first full-screen mode with large item identity, short success feedback and distinct visible error messages. Audio and vibration may supplement text. Support hardware scanners, phone cameras and a controlled manual lookup. Keep Print label and Reprint separate; show printer failure with a retry that cannot create a duplicate package.

Operator and carrier portal

Help owners run today’s work and understand the business without exposing unrelated operators.

O01 Overview

Place actionable exceptions first: departures needing assignment, vehicles blocked, documents needing changes and unsettled items. Each alert shows owner, age, business impact and next action. Below it, show today’s work, capacity utilisation and earnings freshness. Provide branch, service and date filters; retain the selected scope.

O02 Work and departures

Use tabs for Departures, Ride jobs, Shipments and Freight rather than mixing unlike records into one ambiguous table. Departure detail combines manifest, seat map, vehicle, crew, timetable, passenger notifications and changes. Publishing performs readiness checks. Vehicle substitution shows affected seats and customers before committing the replacement.

O03 Fleet and O04 People

Vehicle detail includes ownership, service eligibility, seats, cargo limits, documents, availability calendar, inspections and maintenance. Driver detail shows role, approvals, assignment, shifts and expiry. A blocked vehicle remains discoverable with its reason; hiding it entirely would leave the owner unable to resolve the issue. [5; A, pp. 13-14]

O05 Freight opportunities and agreements

Approved carriers can filter relevant opportunities by route, pickup window, equipment and capacity. Offer submission records amount, included services, validity and conditions. Accepted terms are preserved with the job. If marketplace procurement is not enabled, show assigned contracted work only. [11]

O06 Statements and O07 Organisation settings

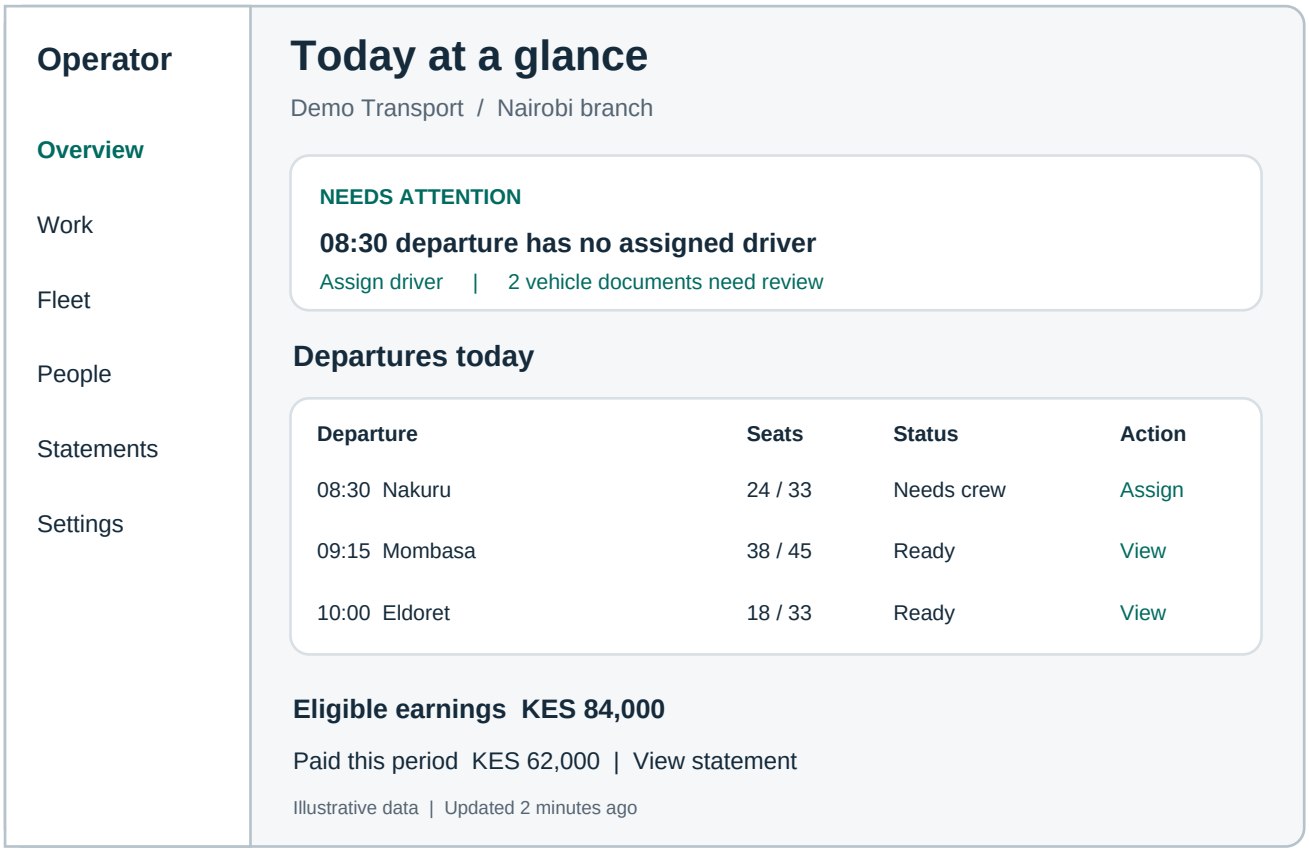
Statement detail traces earned service items through fees, adjustments, eligibility, settlement batches and payout outcomes. Include downloadable statements, dispute reference and reason for withheld items. Payout-account changes and material payouts need separate review. Settings cover branches, service scope, staff permissions and notification recipients. [A, pp. 5, 15, 23]

Small operator usability

Provide a mobile summary with Today, Fleet and Money shortcuts. Keep dense planning tables on desktop. An owner-driver may switch into their assigned driver workspace without sharing the owner’s administrative privileges with other drivers. Publishing, bulk reassignment and financial changes use a preview of affected records before confirmation.

Operator screen concept

Desktop overview with priorities above metrics. Illustrative operational records and amounts.



Interaction notes

The operator selects an organisation and branch before acting. Needs attention is a working queue, not a decorative warning banner. Selecting a departure opens a detail panel with the manifest and readiness checklist; Assign opens an eligibility-filtered vehicle and driver picker.

The separate Money area labels eligible earnings and paid amounts explicitly. A stale report shows its timestamp. On mobile, the table becomes departure cards; secondary columns move into the detail screen rather than shrinking into unreadable text.

Assignment review

Before committing an assignment, show the vehicle, driver, time interval and any relevant capacity checks. If another dispatcher changes availability, explain the conflict and refresh the options. The interface must not show Assigned until the server accepts the allocation. [A, pp. 13, 19]

Dispatch and maintenance workspaces

Treat operational exceptions as assigned work with an owner and a resolution path.

X01 Dispatch board

Use a list and map side by side on desktop. Queue tabs show Unassigned, Ready, Active, At risk and Completed, with service filters. Each card shows promise time, capacity needs, operator, location freshness and reason for risk. Maps are optional for list actions. New requests should not push the selected job out of view.

X02 Assignment and route planning

The assignment panel ranks eligible resources and explains exclusions: conflicting work, maintenance, document expiry, capacity or shift. Show pickup travel and turnaround buffers. Route optimisation is a proposal; a dispatcher reviews stop order, windows and capacity before publishing. Preserve completed stops when revising an active route. [A, p. 13]

X03 Breakdown and reassignment

An incident blocks future allocations and opens a replacement workflow. Select affected passengers and packages, record the old assignment ending, allocate replacement resources and capture physical handover. Customer updates include the revised estimate. Keep the original service promise for reporting, rather than rewriting history to make the delay disappear.

M01 Inspection and M02 Work order

Drivers capture checklist outcomes and evidence. Maintenance staff see priority, vehicle, reported issue, due date, parts, labour and repair evidence. An existing unresolved defect remains visible even after a later checklist passes. Returning a vehicle to service requires the configured authorised sign-off. [12; A, p. 14]

M03 Fleet calendar and operating costs

Show booked intervals, service blocks and inspection deadlines in one calendar. Fuel entries record quantity, cost, odometer and receipt, with anomalies sent for review. Label telematics source and freshness; do not imply that every device provides fuel or diagnostic data. Phone and installed-tracker disagreement should appear as uncertainty, not a smooth invented route.

Operational escalation

Every exception has severity, current owner, next action and due time. Separate urgent safety escalation from ordinary delay support. Notify the assigned team and record acknowledgement. An emergency control must describe the actual available response path; public competitor features do not establish this platform’s response capability. [6; A, p. 21]

Warehouse merchant corporate and recipient views

Extend the same platform with focused interfaces for specialist work.

W01 Warehouse tasks and W02 Inventory

Warehouse staff start with assigned Receive, Put away, Pick, Pack and Dispatch tasks. Every movement identifies item, owner, quantity, unit and location. Picking confirms item and bin; packing compares expected and packed quantities. Stock screens distinguish Available, Reserved, Quarantined and Damaged. Adjustments require reasons and appropriate review. [13; A, p. 12]

W03 Merchant fulfilment

A merchant sees only their stock, orders, delivery performance and returns. Batch imports preview row errors before creating records. Provide replenishment status, shortages, picking progress and shipment references. Customer order numbers and package IDs remain distinct. Returns pass inspection before stock becomes available again.

K01 Corporate policy and approvals

Corporate managers manage employees, cost centres, service permissions, budgets and approval thresholds. An approval request shows traveler, purpose, itinerary, policy exception and expected cost. Separate Approve travel from Pay invoice. Employee removal stops future access while retaining historical invoices. [A, p. 10]

K02 Ride booker and shuttle roster

An authorised booker chooses the traveler, pickup, destination, time and cost centre, with notes for the driver. Show whether the traveler has received the trip details. Recurring shuttles use dated departures and roster exceptions, not a single permanent booking. Changes to recurring trips preview the future dates affected. Bolt’s public Ride Booker documents booking on behalf of a passenger and SMS notification. [7]

R01 Recipient page and R02 Guest traveler

The recipient page works in a mobile browser with an expiring scoped link. Show package count, delivery window, branch instructions, last update and a contact option. Sensitive details and collection actions require the agreed verification. A guest traveler sees their ride identity and pickup instructions without gaining access to the person who paid.

Partner legs

For partner-operated or cross-border services, show the responsible carrier, external reference, next milestone and documents needing attention. Use Partner update received at rather than implying continuous GPS. A handoff failure routes to an internal owner; customers should not have to discover which company is responsible. [A, pp. 3, 11]

Finance support and administration

Make back-office work traceable and understandable to the people affected by it.

F01 Collections and reconciliation

Finance begins with unresolved payments by age and amount. Detail shows order, attempts, provider references, expected versus received amount and reconciliation history. Unknown means not yet resolved, not failed. Offer Check status or Escalate before a retry that could duplicate money movement. [A, p. 15]

F02 Refunds and F03 Settlement

Refund review shows cancellation or claim basis, prior refunds, amount and approval requirement. Settlement lists eligible service items, fees, cash offsets and adjustments with supporting references. A different authorised person approves material payouts. Separate Draft, Awaiting approval, Submitted, Paid, Failed and Unresolved. A timeout never produces an automatic duplicate payout.

S01 Support case

Search by customer-provided booking or shipment reference within permission scope. A case combines the booking, trip, package, payment and communication timelines, with sensitive fields masked. Show current owner, urgency and next response. Separate internal notes from customer-visible messages. A support agent can request an adjustment without necessarily approving it.

S02 Safety and appeals

Provide restricted incident access, evidence handling and escalation acknowledgement. Driver and customer reports both receive a reference and follow-up path. Show an account restriction’s reason category, effect and appeal action where disclosure is appropriate. Do not turn a low rating into an unexplained automatic penalty.

A01 Administration and A02 Audit

Administration covers operator review, service areas, branches, roles, rate cards, document rules and integration health. Use Draft, Review and Publish for changes that affect prices or service availability. Preview the effective date and impacted organisation. Audit captures actor, reason and old and new values. Cross-operator sensitive access requires explicit scope and recorded purpose. [A, pp. 5, 21]

Management reporting

Show route and service trends, utilisation, empty running, pickup punctuality, delivery success and exception age with metric definitions and freshness. Financial totals reconcile to the ledger. Provide scoped exports with personal data minimised. Read-only management access should not imply dispatch or payout authority. [A, p. 23]

Passenger journeys across roles

Cross-role handoffs are the acceptance criteria for a usable frontend.

Scheduled booking through boarding

Step	Visible handoff	Required recovery
1 Search and hold	Customer chooses route segments and seats; server returns hold expiry.	If seats are taken, refresh availability and retain passenger input.
2 Pay and confirm	Customer sees payment progress; operator receives a confirmed booking only after inventory and payment rules pass.	Late payment is recorded; resecure inventory or open rebook or refund handling.
3 Prepare departure	Operator assigns vehicle and crew; conductor downloads the correct manifest.	Block publication or dispatch when readiness fails; explain the blocker.
4 Board	Conductor verifies ticket, departure and segment; customer receives boarding guidance.	Wrong or duplicate ticket gets a specific result and support path.
5 Complete or disrupt	Crew updates departure; customer sees arrival or a revised estimate.	Replacement preserves passengers, seat reassignment and communication history.

On-demand ride through earnings

Customer accepts a versioned quote. Dispatch sends offers to eligible drivers. A driver’s acceptance becomes confirmed only after server arbitration; other offers close clearly. The customer sees driver identity and arrival status. Pickup verification enables trip start. At arrival, completion records the operational event; the payment result and driver earnings eligibility update independently.

If no driver accepts, offer a meaningful alternative and explain any money release process. If a destination changes, show the revised quote before acceptance. If a cancellation occurs, present the policy-based effect and dispute path to both parties. If a device reconnects after reassignment, reject outdated start actions with a clear explanation. [A, pp. 9-10, 15-16]

Charter and corporate variation

Corporate travel adds policy and approval before dispatch. Charter adds staff quotation, deposit and a full-period vehicle reservation. Neither approval nor deposit alone should masquerade as a confirmed pickup unless the required resource checks have completed.

Goods journeys and exception resolution

Preserve the identity of each package and the person responsible at every handover.

Intercity parcel journey

Sender confirms the quote and package details. The driver collects and scans packages with condition evidence. Origin branch receives against a manifest, records discrepancies and sorts to location. Line-haul loading reconciles packages to the assigned vehicle. Destination branch repeats receipt checks. Last-mile delivery or branch collection verifies the recipient and records evidence. The sender sees the same package milestones in a simpler timeline. [A, pp. 11-12]

Freight journey

Shipper requests capacity and loading conditions. Operator submits or accepts an agreed rate. Dispatch reserves resources and publishes the plan. Driver records arrival, collection and delivery milestones and uploads required documents. Proposed additional charges are reviewed against accepted terms. Finance settles only eligible amounts; documents received does not automatically mean payment complete. [11; A, pp. 11, 15]

Exception	Frontend response	Responsible workspace
One package missing	Show delivered and outstanding items separately; open discrepancy with last custody event.	Branch, dispatch and support.
Recipient absent	Record attempt and evidence; propose retry, branch hold or return under policy.	Driver then dispatch.
Damaged package	Record condition, affected item and evidence; prevent silent acceptance.	Branch or driver; claims reviewer.
Wrong scan	Explain expected item or manifest; permit controlled correction with a reason.	Branch supervisor.
Vehicle breakdown	Block resource, create replacement and record passenger or package transfer.	Dispatch and maintenance.
Return approved	Create a linked reverse shipment; show inspection and refund as separate progress.	Support, warehouse and finance.
Partner update missing	Show last known milestone and next internal owner; avoid a false live map.	Partner operations and support.

Correction adds a new event; it does not erase a handover, failed attempt or financial record. Every exception screen should answer what happened, who owns it, what action is available and what the customer has been told. [A, pp. 12, 19]

Visual system and accessibility

Use the same visual language across products, with density suited to each working environment.

Design element	Proposed specification
Colour	Navy #152A3A for primary text; teal #006B60 for primary actions; background #F5F7F9; white surfaces. Validate every actual text and background pair.
Type	Use a readable system sans-serif. Mobile body 16 px; desktop body 14 to 16 px; headings 24 to 32 px. Support platform text scaling.
Spacing	Use a 4 px base with 8, 12, 16, 24 and 32 px spacing. Customer and driver screens use generous vertical separation.
Controls	Prefer 48 px mobile touch targets. Primary actions remain clear above the keyboard and device safe area.
Responsive layout	Below 768 px use single-column cards; 768 to 1199 px use collapsible navigation; 1200 px and above support split list and detail views.
Status	Use label plus icon plus colour. Distinguish Draft, Waiting, Confirmed, Blocked and Needs attention; keep service and payment labels separate.
Tables	Sticky header, meaningful filters, sortable columns, selection count and bulk-action preview. On small screens use record cards or a clearly labelled scroll region.
Maps	Lazy-load maps, show accuracy and update age, and keep a text equivalent for destinations and progress.
Components	Shared location picker, quote summary, seat selector, ticket, timeline, evidence uploader, money breakdown, sync indicator and exception panel.

Accessibility requirements

Target WCAG 2.2 AA for web screens and equivalent native accessibility support. Test keyboard navigation, visible focus, screen-reader labels, error summaries and reflow. Normal text generally needs at least 4.5:1 contrast, with applicable exceptions and large-text criteria. The proposed 48 px touch target is a product choice, not a claim that WCAG requires that size. [14]

Language and local context

Use English and Kiswahili strings throughout core journeys, including errors and receipts. Format KES amounts consistently and display East Africa Time with clear dates. Accept familiar Kenyan phone formats and normalise them. Location capture should combine a pin, stage or branch, landmark and directions. Test translations with actual users rather than translating only navigation labels. [A, p. 6]

State design payments and connectivity

Every screen must distinguish what the user requested from what the server confirmed.

Situation	Suggested message	Available action
M-PESA request sent	Check your phone to authorise the payment.	Change number only through a controlled new attempt.
Payment unresolved	We are checking your payment. Do not pay again yet.	Check status; contact support with reference.
Paid but no seat secured	Payment received. Your seat is not confirmed.	View rebooking or refund case and next step.
Seat hold expired	Your seat reservation expired. Your details are saved.	Choose available seats; show payment state separately.
Offline evidence captured	Saved on this phone. Waiting to send.	View pending uploads; retry when connected.
Sync rejected	This job changed before your update arrived.	Review conflict and contact dispatch; retain evidence.
Tracking stale	Last location update 8 minutes ago.	View latest milestone or contact the assigned team.
Payout unresolved	Payout submitted. Final status is being checked.	View reference and reconciliation status.
Upload interrupted	Photo saved. Upload incomplete.	Retry upload without repeating delivery.
Authorisation removed	Your access to this workspace has changed.	Switch workspace or contact administrator.

Offline and privacy behaviour

Encrypt cached field data and minimise its lifetime. Pending actions retain unique event IDs and are removed only after acknowledgement. Show accepted, duplicate and rejected outcomes without making users decipher technical codes. On shared devices, prevent the next user from seeing the previous person’s cached manifests or documents. [A, pp. 16, 21]

Prevent misleading completion

Maintain three separate dimensions: booking or commercial state, trip or delivery state, and payment state. A delivered shipment may still have unresolved payment; a cancelled ticket may already be paid. Disable repeated submission while checking the authoritative result, but retain a visible recovery route when a request stalls. [A, pp. 15, 19]

Permissions communication and trust

Design permissions into the workflow and enforce them on the server.

Action	Can initiate	Additional control
Publish a departure	Authorised operator staff.	Readiness, vehicle and inventory checks.
Assign or reassign	Dispatcher within assigned scope.	Eligibility and conflicting-allocation checks.
Board or deliver	Assigned conductor or driver.	Departure, package and evidence validation.
Correct custody	Permitted supervisor.	New reasoned event; original retained.
Request refund	Customer through case; authorised staff.	Policy checks and reviewer when required.
Approve payout	Separate finance approver.	No self-approval; final provider result tracked.
Release blocked vehicle	Authorised maintenance reviewer.	Repair verification and recorded sign-off.
Read sensitive incident	Assigned safety staff.	Limited evidence access and audit.
Change rates or roles	Authorised administrator.	Preview, effective date and audit history.

Notification design

Notify the right participant at booking confirmation, driver assignment, material delay, ready-for-collection, delivery, refund decision and payout outcome. Use in-app updates, push and configured SMS fallback with deduplication. Each notification opens the relevant record and explains the next action. A failed SMS never invalidates a confirmed booking. [A, pp. 7-8]

Protect both sides of the service

Show drivers when work tracking is active and who can see it. Limit contact access to assigned work; use masked communication where the provider supports it. Customers and recipients see only relevant route information. Provide separate urgent reporting and ordinary feedback, with accessible complaint and appeal paths. [2, 6; A, pp. 7, 21]

Avoid false scarcity, hidden fees, forced ratings, misleading countdowns and unexplained earnings deductions. Show cancellation consequences before the final action. Support-hours and emergency-response wording must reflect actual staffing and integration. Confirm operating permissions, financial responsibility and applicable legal requirements before production; this specification does not determine those obligations.

Build sequence and acceptance criteria

Preserve the full design scope while releasing complete, testable service journeys.

Recommended delivery sequence

Release 1 establishes identity, organisation scope, design components, fleet readiness, payments, support and a scheduled-passenger pilot. Include customer booking, operator departures, conductor boarding, driver trip detail and finance reconciliation together. Release 2 adds on-demand rides and direct courier delivery with offer arbitration and evidence. Release 3 adds hubs, freight, charter, corporate and merchant workflows. Release 4 extends warehouse depth and partner legs as operational readiness allows. This is a proposed prioritisation, not a delivery estimate.

Frontend implementation boundaries

The architecture proposes Flutter for customer and driver apps and React with TypeScript for web portals. Reuse design tokens, API schemas and terminology across those surfaces rather than forcing every screen into identical code. Keep the existing backend implementation unless a separate review justifies changing it; the PDF’s technology choices are proposals. [A, p. 24]

Frontend actions consume server-authorised state and allowed operations. Keep provider credentials off devices. Use stable request IDs for retryable actions, server-based hold expiry, and explicit version-conflict handling. Push or live updates should fall back to refresh or polling without claiming that old data is live. Screen IDs in this document can become design frames and backlog references.

Required pilot scenarios

• Two customers attempt the last seat; one succeeds and the other keeps their entered details.

• Payment succeeds after expiry; no false ticket is issued and the resolution is visible.

• Two drivers accept one offer; both receive an unambiguous result.

• A driver records evidence offline, restarts the app and syncs twice without duplicate completion.

• One of three parcels is missing; only the delivered items close.

• A vehicle breaks down; replacement, custody and customer updates remain traceable.

• A user changes an organisation or record URL; unauthorised data stays inaccessible.

• Keyboard, screen reader, enlarged text, Kiswahili and poor-network journeys remain usable.

Validate with the community

Run moderated tasks with customers, drivers, small and larger operators, conductors, branch staff, dispatch, finance and support. Measure booking completion, pickup clarification, dispatch time, scan errors, unresolved sync age, payment exception age and payout comprehension. Set targets after baseline sessions; competitor marketing figures are not performance promises. Resolve launch-service failures before expanding service coverage. [A, pp. 23, 25-27]

Sources for mobility and operators

Public sources consulted 17 October 2026. Numbered references identify the evidence behind findings; proposed layouts and workflows are design recommendations.

A Kenya Transport and Logistics Architecture

User-provided architecture, version 1.0, prepared 18 September 2026. Pages 3-6 establish scope and roles; 9-16 define workflows; 19 defines state separation; 21-27 cover controls, implementation and acceptance.

1 Uber guide to using the app

Rider request, pickup, identity, prices and regional feature caveat.

Open Uber guide to using the app

2 Uber Verify Your Ride

Optional PIN verification and trip-start behaviour; public UK help page.

Open Uber Verify Your Ride

3 Uber calculation of prices

Public driver guidance on upfront earnings and excluded adjustments; market-specific guidance.

Open Uber calculation of prices

4 Uber driver earnings

US public explanation of earnings summaries, statements and fare breakdowns.

Open Uber driver earnings

5 Bolt Fleet Portal

Driver and vehicle management, reports, invoices and payouts.

Open Bolt Fleet Portal

6 Bolt safety features

Kenya-localised public page describing sharing, pickup codes and safety tools, with availability caveat.

Open Bolt safety features

7 Bolt Business booking guide

Booking for another passenger, schedule, expense notes and SMS details.

Open Bolt Business booking guide

Sources for logistics and accessibility

Public sources consulted 17 October 2026. Numbered references identify the evidence behind findings; proposed layouts and workflows are design recommendations.

8 BuuPass booking guide

Official guide to bus booking and payment options.

Open BuuPass booking guide

9 Onfleet delivery platform overview

Driver navigation, evidence, recipient tracking and communication.

Open Onfleet delivery platform overview

10 Onfleet barcode scanning

Pickup and drop-off barcode workflow in official developer documentation.

Open Onfleet barcode scanning

11 Uber Freight carrier portal

Load search filters, status and proof-of-delivery uploads.

Open Uber Freight carrier portal

12 Fleetio inspections in the mobile app

Inspection failures, open issues and issue resolution behaviour.

Open Fleetio inspections in the mobile app

13 Odoo barcode receipts and deliveries

Official version 17 documentation for scanning transfer operations.

Open Odoo barcode receipts and deliveries

14 W3C Web Content Accessibility Guidelines

WCAG 2.2 reference for contrast, focus, target sizing and accessible interaction.

Open W3C Web Content Accessibility Guidelines